

Scott J. Gilbert

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Aspiring computer engineer with strong foundations in hardware design, embedded systems, and full-stack development. Skilled in C/C++, Python, systems engineering, low-level programming, and data processing, with hands-on experience in real-time systems, machine learning, and DevOps.

Education

University of Illinois Urbana-Champaign

Bachelor of Science in Computer Engineering
Minor in Business

May 2029 | GPA: 4.00/4.00 (Dean's List) | Chancellor's Scholar

The Grainger College of Engineering
Gies College of Business

Activities and Societies: Campus Honors Program, Illini Solar Car, Project: Code, Illini Redstone Computing, Engineers without Borders, Marching Illini

Relevant Coursework: Differential Equations, Computational Linear Algebra, Computers & Programming, Quantum Physics, Thermodynamics, Discrete Mathematics, Data Structures, Analog Signal Processing, Fundamentals of Accounting

Professional Experience

Computer Administrator

Oct 2025 - Present

University of Illinois Campus Honors Program (CHP)

Urbana, Illinois

- Support core CHP operations by maintaining hardware infrastructure, external storage integrations, and web services across various providers.
- Coordinate with 15+ CHP staff, IT staff, and other administrators to modernize computing resources and increase accessibility/utility to students.
- Synchronize data between university admissions and CHP databases, reducing entry time for critical record processing by up to two weeks.

Full-Stack/AI Intern

June 2025 - Aug 2025

Team2Go Inc.

Yongin-si, Gyeonggi-do, South Korea (Remote)

- Collaborated with my intern team to deploy a full stack AI application publishing platform supplemented by Retrieval-Augmented Generation, a custom chat interface, rigid boundary/instruction declaration, easy spin-up/spin-down in Docker, and full localization in English and Korean.
- Explored and deployed psychology techniques to reduce LLM hallucination and improve response accuracy without sacrificing ease of deployment.
- Reported regular progress to senior executives and shipped key deliverables two weeks ahead of schedule while exploring LLM design/psychology.

Leadership/Involvement

Array/Maximum Power Point Tracking (MPPT) Lead & Webmaster

May 2026 - Present

Illini Solar Car

Champaign-Urbana, Illinois

- Develop MPPT optimization/communication firmware, evaluate solar array iterations with OpenGL tools, and manage budget of ~\$45,000.
- Serve on the executive board, maintaining mission-critical software (MediaWiki knowledgebase, public website/outreach platform, etc.) across hosting services, managing associated virtual machines, DNS, and proxies, and addressing critical failures and other service requests while on call.

Backend Team Lead

Sep 2025 - Dec 2025

Project: Code UIUC (Agri-Sense Project)

Champaign-Urbana, Illinois

- Managed team of ten developers focused on deploying data handling, APIs, and core logic for a data-driven, all-in-one crop monitoring platform.
- Oversaw integration of UART, OneWire, and I2C data buses on IoT boards to communicate sensor readings and other metadata to central backend.
- Maintained production branches, versioning and CI policies, and shared progress/API configuration with project and hardware/frontend leads.

Eagle Scout

Mar 2018 - Dec 2024

Scouting America

Hoffman Estates, Illinois

- Organized sponsorship operations, engineering and construction management, and workplace operations for a coordinated nature restoration effort where volunteers contributed over 100 working hours with no safety incidents.
- Earned 27 merit badges, 10+ other certifications/awards, and held 5 leadership positions, including troop guide and assistant senior patrol leader.
- Practiced and continue to practice Scout Oath and the 12 points of the Scout Law in service to the community and to the nation.

Projects

Illini Redstone Computing (Engineering Club) Infrastructure

Jan 2026 - Present

- Designed IRC's computing backbone, deploying three container orchestrations and various other services across four different hosting providers.
- Integrated gaming servers, WebDAV, DNS, web-based administration, datastores, messaging services, staff/admin authentication, API gateways, and secure connection tunneling into the full IRC stack, keeping services modular to support future upgrades/expansions.

Delos (Solar Electric Vehicle)

Oct 2025 - Present

- Modernized firmware to resolve five-year-old systemic blocking issues, implementing boost converter power delivery and CAN messaging.
- Helped design ISC's most efficient solar array (1.5 kW, 6 m²) and finalized/deployed key wheel, dash, and MPPT subsystems that powered the car for 650+ miles without any failures, all within one month's turnaround from the start of electrical integration.
- Contributed to development and installation of the car's electrical/computer systems, including enclosures, cooling, wiring, and ergonomics.

Skills

- **Programming:** C, C++, Python, SQL, Java, Assembly, TypeScript, PHP, Wolfram
- **Datastores:** MySQL, SQLite, MariaDB, PostgreSQL, pgvector, S3, UploadThing, Backblaze, CSV
- **Numerical Analysis:** Jupyter Notebook, NumPy, SciPy, Matplotlib, Mathematica, Excel, Google Sheets
- **Embedded Systems/Microcontrollers:** Arduino, ELEGOO, CAN, UART, I2C, OneWire, KiCad, NXP, Arm Cortex, AMD Vivado
- **Industry Tools:** Docker, Podman, MCUXpresso, LTSpice, Git, GitHub, Visual Studio Code, JetBrains, Google Workspace, Microsoft Office, Slack
- **Workflow Automation & CI/CD:** Vercel, Claude Code, GitHub Copilot, GitHub Actions, n8n, Make
- **Technical Expertise:** Embedded Systems, Firmware Design, Backend Development, PCB Design, AI/ML Application, Virtualization/Containerization

Awards: National Merit Scholarship Recipient, AP Scholar with Distinction, Gold President's Volunteer Service Award, Seal of Biliteracy (Mandarin)